Budgets as grades: which equal-meaning inputs a budgeted reasoner distinguishes

Jinu Jang
September 2026 (draft v0.1)

Abstract

A recognizer that answers Horn entailment queries with a fixed computation budget does not treat logically identical inputs alike. We ask which of them it separates, and answer with a graded object. The closure operator of propositional Horn logic factors into stages T_k, the atoms derivable in k parallel rounds, with T_0 = id, $T_j \circ T_k = T_{j+k}$, and limit the closure; we prove in Lean 4 (core only) that the limit is sound and complete, that budget-k behavioural identity is blind to order and repetition at every grade, and that a trace is separated from a redundant extension at budget k exactly when the extension moves some derivation across the budget. This yields a preregistered prediction: a derivable clause that shortens the target derivation changes a budgeted recognizer's decision at a low budget and not at a high one, while an equally redundant clause that preserves depth changes nothing. On 200 fresh cases the prediction held for a learned iterative reasoner (interaction 0.885, 95% CI [0.84, 0.93]), whose decisions coincided with the k-round symbolic reasoner at every budget. A second prediction, the composition law read on inputs ("a hint buys exactly its depth"), was then tested on the same weights and failed at tight budgets (agreement 0.90 and 0.83 against a 0.95 criterion) while holding for a model trained under a smaller budget. Two language models (0.5B, 1.5B) did not read the table at the decision level; at the margin level their sensitivity to redundant clauses was dominated by lexical overlap with the query, with a small residual in the depth direction. The graded structure therefore describes what a budgeted reasoner distinguishes; whether it also obeys the structure's composition law is a property of training, not architecture, and was decided by testing the law rather than assuming it.

1. Question

Two inputs with the same meaning are, to a recognizer, two inputs. The recognition-paths framework [@rp] makes this precise for Horn logic: a recognizer ρ induces a behavioural identity $\approx_\rho$ on premise traces, logical identity $\equiv_L$ is theory equivalence, and ρ factors through $\equiv_L$ if and only if it identifies every pair of logically identical traces (the Recognition Factorization Theorem). The empirical part of that programme found that no recognizer examined, language models to 1.5B and small constructed models alike, identifies logically identical traces; in particular a trace and its extension by a *derivable* clause, which changes no consequence, are distinguished by every one of them [@ppi].

That is a negative result about identification. This paper asks the positive question it leaves open: *which* logically identical inputs does a recognizer separate, and is there a mathematical object that predicts the answer? The object we propose is the graded closure: forward chaining stopped after k rounds. Its grades are computation budgets, and it makes two predictions that can be preregistered and tested, one about identification and one about composition. The first held. The second failed for the model it was primarily tested on. Both outcomes are the content of the paper.

2. The graded closure

Fix a set of atoms and a Horn theory Γ (a set of clauses $a_1 \wedge \ldots \wedge a_m \to b_1 \wedge \ldots \wedge b_n$). For a set of atoms S let $\mathrm{step}_\Gamma(S)$ be S together with the heads of all clauses whose bodies lie in S, and let

$$T_0 S = S, \qquad T_{k+1} S = \mathrm{step}_\Gamma(T_k S).$$

The following are proved in `RecognitionPaths/Graded.lean` (Lean 4, no Mathlib; names in parentheses):

1. $T_j(T_k S) = T_{j+k} S$ (`rounds_add`): the grades add.
2. $S \subseteq T_k S \subseteq T_{k+d} S$ (`rounds_extensive`, `rounds_le_add`); T_k is monotone in S and in Γ (`rounds_mono`, `rounds_mono_theory`).
3. Soundness: every atom in $T_k S$ holds in every model of Γ containing S (`rounds_sound`). Completeness: $T_\infty S = \bigcup_k T_k S$ is a model of Γ (`limit_models`), hence semantic entailment is entailment at some finite grade (`entails_iff_exists_rounds`).

So $(T_k)_{k \in \mathbb{N}}$ is an $\mathbb{N}$-graded monad on the poset of atom sets whose ∞-stage is the (idempotent) closure monad. Only the limit is idempotent: $T_k \circ T_k = T_{2k} \neq T_k$.

Graded entailment and identity. Write $\Gamma \vdash_k (H \Rightarrow g)$ for $g \in T_k H$. Budget-k identity of two traces u, w is

$$u \approx_k w \Leftrightarrow \forall q,\ \Gamma(u) \vdash_k q \leftrightarrow \Gamma(w) \vdash_k q$$

(`GradedEquiv`). It is an equivalence relation, blind to permutation and repetition at every grade (`gradedEquiv_of_perm`, `gradedEquiv_dup`), and identity at every grade implies $\equiv_L$ (`logicalEquiv_of_gradedEquiv`). The converse fails at any fixed grade, and the failure is characterised exactly:

> **Theorem (`gradedEquiv_append_iff`).** For a trace w and a clause c, $w \approx_k w + [c]$ if and only if every query answered within k rounds from $w + [c]$ is answered within k rounds from w. Equivalently (`not_gradedEquiv_append_iff`), w and $w + [c]$ are separated at budget k iff some query's derivation is moved by c from depth $> k$ to depth $\leq k$.

For a derivable c the separation is transient: for every query there is a grade beyond which w and $w + [c]$ agree (`gradedEquiv_append_derivable_eventually`).

The budgeted recognizer. $\rho_k(w, q) = [\Gamma(w) \vdash_k q]$ is a theory-factoring recognizer; its behavioural identity is $\approx_k$ in every context (`roundsRecognizer_contextEquiv_iff`). The composition law read on inputs (`entailsK_presaturate`) says

$$\rho_k(w,\ T_j\{a\} \Rightarrow g) = \rho_{j+k}(w,\ \{a\} \Rightarrow g):$$

pre-saturating the hypothesis by j rounds and reading with budget k is reading with budget $j + k$. A hint buys exactly its depth.

The mathematics here is elementary. Its role is to name the object whose equations become predictions; the paper's claims are about which recognizers satisfy them.

3. Two predictions

P1 (identification; RQ2). Take a base theory D with a target query of depth $d \geq 3$ from a single hypothesis atom. Let F add a derivable clause that shortens the target's depth by the least possible amount, and C add a derivable clause that preserves it; $D \equiv_L F \equiv_L C$. For a recognizer with budget k let $\mathrm{dis}_X(k)$ be the fraction of cases whose target decision differs between D and X, and $\Delta(k) = \mathrm{dis}_F(k) - \mathrm{dis}_C(k)$. The theorem predicts $\Delta(k) > 0$ for k below the base depth and $\Delta(k) = 0$ above it. The preregistered test is the interaction $I = \Delta(2) - \Delta(4)$ for a learned reasoner trained at four rounds: P1 holds iff $I > 0$ with a paired case-bootstrap 95% CI excluding 0 and $\Delta(2) \geq 0.10$.

P3 (composition; RQ2b). For the same weights, with hypothesis sets $H_j = T_j \{a\}$, the agreement rate between $\rho(H_j; k)$ and $\rho(H_0; j + k)$ must have CI lower bound ≥ 0.95 on the four pairs $(j, k) \in \{(1, 2), (2, 1), (2, 2), (1, 3)\}$ where the symbolic yes-rate changes between budgets k and $j + k$ (so that a constant answer cannot pass).

Both were committed with their exclusion rules, tie handling, sample-size rationale, controls and failure sentences before any learned recognizer was run (`docs/PREREGISTRATION_RQ2.md`, `docs/PREREGISTRATION_RQ2B.md`).

4. Instruments

Table. 7 atoms; 200 base theories of 5 random clauses with a target of depth ≥ 3 from atom a (depth 3: 177, depth 4: 23), generated from one seed and taken in order after fixed exclusions; conditions D, F (minimal shortening; never the direct clause), F_1 (the direct clause $a \rightarrow$ goal, maximal shortening and maximal lexical overlap with the query), C (depth-preserving, overlap with $\{a, \text{goal}\}$ matched to F in 181/200 cases), and L (one clause deleted so the target becomes underivable: the logic-change control). Four queries per case: the target t, a depth-1 atom d_1, a non-derivable atom n, and the reversed query r. Logical identity of D, F, F_1, C is certified by closure equality on every single-atom hypothesis. The RQ2b table takes the 200 D theories with hypothesis sets H_0, H_1, H_2 for t and n.

Recognizers. (i) The exact closure oracle and the k-round symbolic reasoner, $k \in \{1, 2, 3, 4, 6\}$ (control validation: the theorem holds for them by construction). (ii) A constant-NO recognizer and Pythia-70M (non-reading controls). (iii) The iterative learned reasoner: atom states updated along clauses by a message MLP and a GRU cell, max-aggregated per head atom (so permutation- and repetition-invariant), read out from the goal and hypothesis states; the number of rounds is the budget. 75k parameters, trained 6000 steps on random 2–6-clause theories over 7 atoms with the 1000 evaluation theories excluded up to atom relabelling, at 4 rounds (primary) and at 2 rounds (secondary). (iv) A 7-atom set recognizer (max-pooled clause encoder; no budget knob). (v) Qwen2.5-0.5B-Instruct and 1.5B-Instruct, raw prompt, one forward pass, decision by the YES/NO logit comparison, on CPU.

Decisions are Boolean with ties to NO; the extended observation is the real margin. All materials are frozen with SHA-256 locks and cited by commit (see `REPRODUCE.md`).

5. Results

5.1 P1 holds, and the learned reasoner is the symbolic one

budget k	**acc(D, target)**	dis$_F$	dis$_{F_1}$	dis$_C$	dis$_L$	**Δ [95% CI]**
1	0.00	0.01	1.00	0.00	0.00	0.01 [0.00, 0.01]
2	0.00	**0.89**	1.00	**0.00**	0.00	**0.89 [0.84, 0.93]**
3	0.89	0.11	0.11	0.01	0.89	0.10 [0.06, 0.15]
4	1.00	**0.00**	0.00	**0.00**	1.00	**0.00 [0.00, 0.00]**

budget k	acc(D, target)	dis_F	dis_{F_1}	dis_C	dis_L	Δ [95% CI]
6	1.00	0.00	0.00	0.00	1.00	0.00 [0.00, 0.00]

Table 1: the 4-round learned reasoner on 200 cases. $I = \Delta(2) - \Delta(4) = 0.885$, 95% CI [0.84, 0.93]; P1 holds.

The learned reasoner's decision table coincides with the k-round symbolic reasoner's at every budget: at $k = 2$ the 177 depth-3 bases are NO and their F extensions (depth 2) YES, the 23 depth-4 bases NO on both; at $k = 4$ everything is YES and only L is separated. Every D–F disagreement is F correct and D wrong. The direct clause F_1 separates at $k = 1$ (it is read as a depth-1 derivation) and not at $k = 4$, so lexical overlap plays no role for this recognizer. The controls behave as required: the oracle and the constant recognizer give $\Delta = 0$, Pythia-70M gives $\Delta = -0.02$ [−0.04, 0.01] with $\mathrm{dis}_L = 0.04$ (it does not read).

A caveat stated in advance: in this architecture information flows one hop per round, so a *correct* model must show this pattern. The empirical content is that training produced this reader and not a shortcut (in-distribution accuracy 0.996; no clause-count or lexical heuristic survives at $k = 2$), that the gap closes exactly at $k = 4$, and that the pattern is what the theorem says and nothing more.

The 2-round model (trained with a budget too small for its data, validation accuracy 0.960) shows the same pattern, $I = 0.665$ [0.60, 0.74], and closes the gap at $k = 4$ although it never ran four rounds in training. It differs in one informative way: at its own budget it says YES on 16% of the depth-3 bases it cannot derive and its decision moves on 10% of the depth-preserving extensions C. A model trained under an insufficient budget acquires a shortcut component that a merely redundant clause can trigger; depth accounts for 0.67 of its 0.77 F effect.

The set recognizer (no budget) identifies almost every extension at the decision level ($\mathrm{dis}_F = 0.01$, $\mathrm{dis}_C = 0.00$) and reads the logic change ($\mathrm{dis}_L = 0.94$). Its margins order the conditions $F_1 > F > C > 0 > L$ (mean shifts +1.91, +0.65, +0.23, −9.3 logits): a recognizer without rounds still orders redundant extensions by depth shortening in its extended observation.

5.2 P3 fails for the primary model

pair	symbolic	4-round model (primary)	2-round model
$j = 1, k = 2$	1.000	**0.897 [0.870, 0.925]**	0.990 [0.980, 0.998]
$j = 2, k = 1$	1.000	**0.830 [0.797, 0.863]**	0.975 [0.960, 0.990]
$j = 2, k = 2$	1.000	0.995 [0.988, 1.000]	0.998 [0.993, 1.000]
$j = 1, k = 3$	1.000	0.983 [0.970, 0.995]	1.000 [1.000, 1.000]

Table 2: agreement between $\rho(H_j\,; k)$ and $\rho(H_0\,; j + k)$ on 200 cases × 2 queries.

Two of the four primary pairs fall below the criterion; P3 fails. As preregistered: *on 200 fresh Horn cases the learned reasoner did not satisfy the composition law $T_k \circ T_j = T_{j+k}$ out of its training format; the graded-monad description fits its budget behaviour but not its hypothesis-set behaviour, and the law is a property of the symbolic reasoner that the learned one does not inherit.*

The failure has one direction: in all 41 and 68 disagreements the model answers NO from the pre-saturated hypotheses at the tight budget where the law says YES. It reads multi-atom hypothesis sets correctly given slack (accuracy 1.000 at $k = 4$ with H_1 and H_2), agreement falls with the size of the hint (at $j = 2, k = 1$:

$|H_2|$ = 3: 0.73, 4: 0.63, 5: 0.55) and is near-perfect on depth-4 cases (0.957), where a spare round exists. The learned operator satisfies $T_k \circ T_j \subseteq T_{j+k}$ with strict inclusion on 20–35% of cases at the tight budgets: it loses part of a round when the derivation starts from a set rather than an atom. The 2-round model, which had to propagate from whatever it was given within two rounds, passes all four pairs; this was not preregistered for it and is reported as an observation.

5.3 Language models: no decision-level reading; margins follow overlap

On this rendering (7 atoms, 5–6 clauses) Qwen2.5-0.5B answers YES on every query of every condition and Qwen2.5-1.5B answers NO on every query, so the decision-level test cannot be evaluated for either (the audit protocol's Gate R). Their margins carry partial reading (comparative accuracy $[m_t > m_n]$: 0.79 and 0.89). The mean margin shift on the target relative to D, with paired-bootstrap CIs, is:

recognizer	$F - C$	$F_1 - F$	L
Pythia-70M (non-reader)	0.010 [0.005, 0.014]	0.004 [−0.001, 0.009]	−0.080
Qwen2.5-0.5B	0.033 [0.026, 0.040]	0.098 [0.091, 0.104]	−0.406
Qwen2.5-1.5B	0.048 [0.030, 0.067]	0.285 [0.268, 0.302]	−0.207
set recognizer	0.417 [0.289, 0.540]	1.264 [1.124, 1.407]	−9.32

Table 3: signed margin shifts (logits), 200 cases; exploratory.

The shortening clause moves the LLM margins more than the depth-preserving clause, with CIs excluding 0, but by 0.03–0.05 logits: three to five times the non-reader's surface residual (which exists because F mentions the hypothesis atom in 198/200 cases and C in 131/200; on the 181 overlap-matched cases $F - C$ is 0.025 and 0.029) and an order of magnitude below the direct clause's effect. For these language models, what makes a redundant clause distinguishable is overwhelmingly its lexical overlap with the query; the depth direction is a small residual.

6. What was learned

1. **The graded closure predicts identification.** For a budgeted reasoner, a redundant extension is distinguishable exactly when it moves a derivation across the budget; this held on fresh cases with the full predicted shape (large at k = 2, zero at k = 4, zero for depth-preserving clauses at every k).
2. **The same object's composition law is not inherited by training.** The 4-round model is exactly graded in what it separates but not exactly graded in how it composes hints with budget; a model trained under a tighter budget is. Architecture makes the law available; training decides whether it holds. This is the first equation of the candidate monad to be *refuted* for a constructed recognizer, and it was refuted by a preregistered test of the law.
3. **For the language models examined, depth is not the variable.** The quantity that predicts their (margin-level) sensitivity to redundant clauses is lexical overlap with the query. Whether a language model with a genuine budget knob (chain-of-thought length) obeys the composition law is the natural next test and is not attempted here.

7. Limits and non-claims

Two language models, both under 2B parameters, one rendering, single forward pass; the decision-level LLM tests are not evaluable at this scale and are reported as such, not as refutations. The constructed models are small and the iterative reasoner's architecture makes P1's pattern available to a correct model; the paper's claim is that training produced that model and that the graded description has an equation the model fails, not that the model "has" a monad. Nothing is claimed about the algebraic structure of language models. The Lean development is elementary; it fixes definitions and rules out ambiguity in what was tested, and is not offered as new mathematics.

References

- [@rp] J. Jang, *Recognition paths: logical and behavioural identity for Horn traces* (Lean 4). Zenodo, doi:10.5281/zenodo.22495808.
- [@ppi] J. Jang, *Proof-path invariance: Hankel tables for recognizer identity* (data, code, results). Zenodo, doi:10.5281/zenodo.22495706; draft `docs/DRAFT.md`.
- Models: `jinu0633/recognition-paths-recognizers`, folder `rq2/` (Hugging Face).
- Related work on premise-order sensitivity, set-invariant encoders and graded monads is discussed in `proof-path-invariance/docs/RELATED_WORK.md` and `docs/NOVELTY.md`; the graded-monad notion follows Smirnov (2008) and Fujii, Katsumata and Melliès (2016).